

Student Performance Report

Student: Student row 0 | Generated: 30 July 2026 | Horizons run: E1, E2, E3

50% failure probability	50% pass probability	29% confidence (Low)	HIGH risk level
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Likely range 47-53%. This student sits almost exactly on the pass/fail boundary, so the verdict could flip on a small change. Treat it as uncertain, not as a decision.

Overall summary

The student is at risk of failing.

Strongest positive indicators

- Parental support - 3 = High (Good). 51% of students at this level pass, against a 47% cohort rate (+4 pts, n=697).

Major concerns

- Weekly study time - 6.9 (Needs improvement). 35th percentile of its cohort; students who pass typically reach 10.55.

Immediate intervention: **WITHIN ONE MONTH**

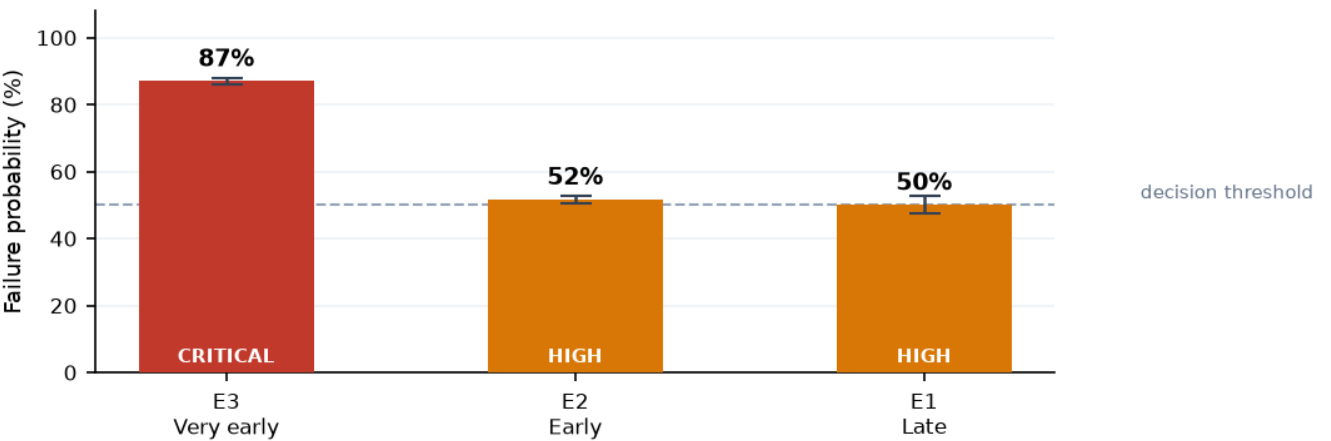
Put a support plan in place this month and set a review date.

Read this first

- ! This dataset merges five separate studies, so each answer is graded against the study it came from rather than the whole file. See Appendix B.

Prediction results

Three models are trained on progressively more information. E3 is available on day one but knows least; E1 is the most accurate but arrives late.



Model	Stage	Data it uses	Fail	Pass	Likely range	Confidence	Risk	Verdict
E3	Very early	Start of term - enrolment and background data only.	87.0%	13.0%	86-88%	High (89%)	CRITICAL	AT RISK
E2	Early	Mid-term - background plus engagement and behaviour.	51.7%	48.3%	51-53%	Moderate (60%)	HIGH	AT RISK
E1	Late	Late term - everything above plus actual grades.	50.0%	50.0%	47-53%	Low (30%)	HIGH	AT RISK

How to read confidence: it combines how far the probability sits from the 50% threshold, how stable the prediction was across the model's trees, how typical this student is of the training data, and how many inputs were supplied.

E3 - Very early model

Start of term - enrolment and background data only.

AT RISK - 87.0% fail / 13.0% pass (risk **CRITICAL**)

Confidence High (89%). Likely range 86-88%.

Clear margin, stable across the ensemble, and typical of the training cohort.

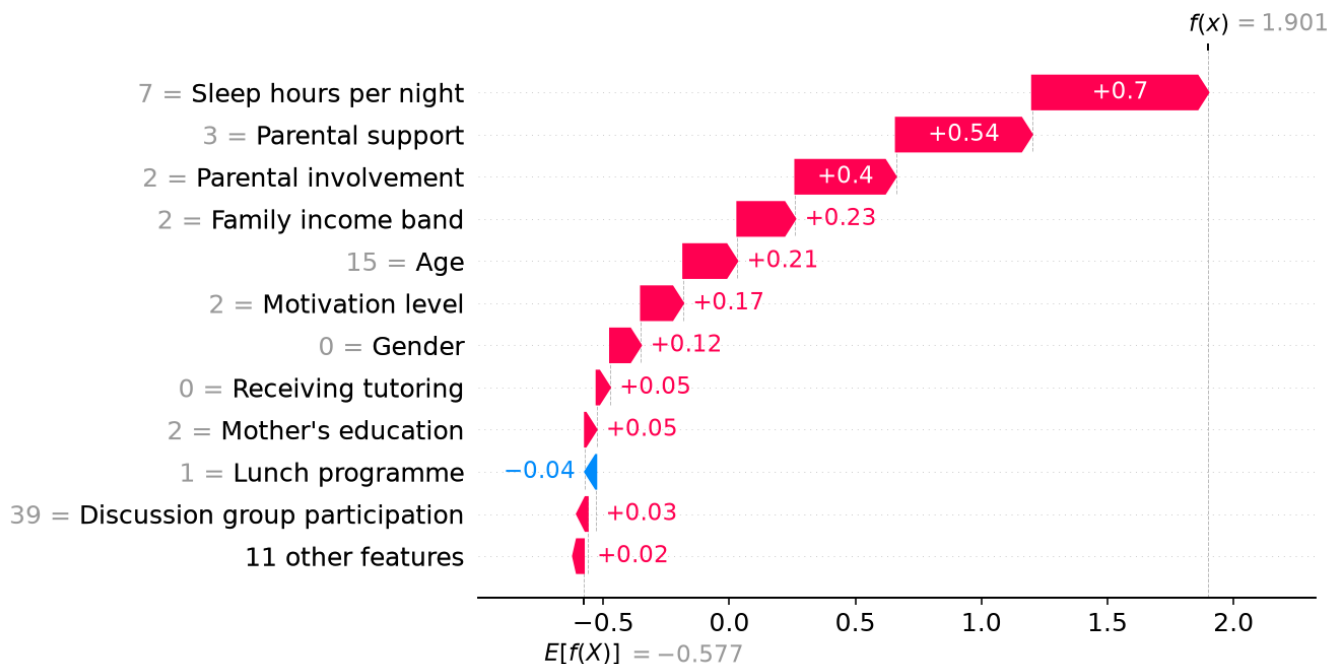


Predicted AT RISK of failing - 87% failure probability (13% chance of passing). Risk level **CRITICAL**. Main drivers: Sleep hours per night; Parental support; Parental involvement. Confidence: High (89/100); likely range 86%-88%. Highest-impact change: Parental support: change from 3 = High to 4 = Very high.

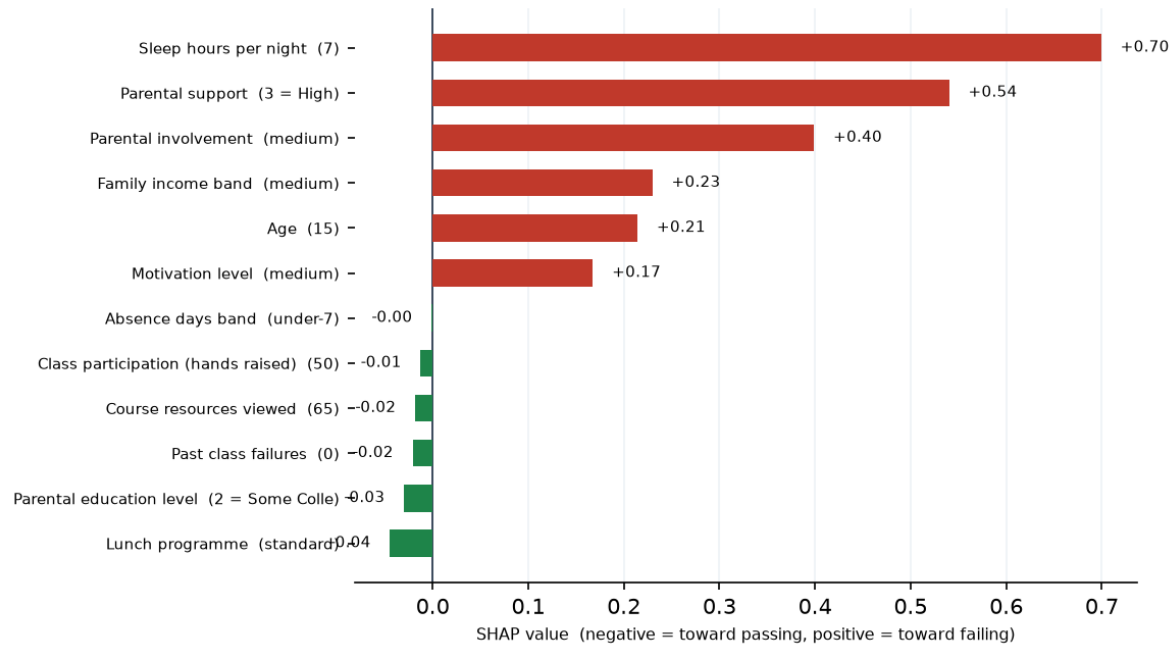
SHAP explainability

Each bar is one answer pushing the prediction away from the cohort average. Red pushes toward failing, blue toward passing; length is the size of the push.

E3 (Very early) - how each answer moved the prediction



E3 - top contributors



Read it like this: Sleep hours per night contributed +0.70 toward predicting failure for this student.

E3 - student profile analysis

Status compares this student against students in the same source study who actually passed. Influence is how strongly the model used the answer for this student. The two can disagree: a good value can still be the model's main worry if everything else is stronger.

Feature	Value	Ideal	Status	Influence	Effect on this prediction
Sleep hours per night	7	-	Not recorded	High	raises risk (+0.70)
Parental support	3 = High	4 = Very high	Good	High	raises risk (+0.54)
Parental involvement	medium	-	Not recorded	Medium	raises risk (+0.40)
Family income band (background)	medium	-	Background	Medium	raises risk (+0.23)
Age (background)	15	15 or below	Background	Medium	raises risk (+0.21)
Motivation level	medium	-	Not recorded	Medium	raises risk (+0.17)
Gender (background)	female	male	Background	Low	raises risk (+0.12)
Receiving tutoring	0 = No	1 = Yes	Moderate	Low	raises risk (+0.05)
Mother's education (background)	2 = 5th to 9th grade	-	Background	Low	raises risk (+0.05)
Lunch programme (background)	standard	-	Background	Low	lowers risk (-0.04)
Discussion group participation	39	-	Not recorded	Low	raises risk (+0.03)
Test preparation course	none	-	Not recorded	Low	raises risk (+0.03)
Internet access at home	yes	-	Not recorded	Low	raises risk (+0.03)
Parental education level (background)	2 = Some College	1 = High School	Background	Low	lowers risk (-0.03)
Past class failures	0	-	Not recorded	Low	lowers risk (-0.02)
Course resources viewed	65	-	Not recorded	Low	lowers risk (-0.02)
Class participation (hands raised)	50	-	Not recorded	Low	lowers risk (-0.01)
Announcements viewed	39	-	Not recorded	Low	raises risk (+0.01)
Course topic (background)	it	-	Background	Low	raises risk (+0.01)
Father's education (background)	2 = 5th to 9th grade	-	Background	Low	raises risk (+0.01)
Education stage (background)	middleschool	-	Background	Low	raises risk (+0.00)
Absence days band	under-7	-	Not recorded	Low	lowers risk (-0.00)

Strengths

Excellent areas

None on this horizon.

Good areas

At or above the typical passing student.

- Parental support - 3 = High. 51% of students at this level pass, against a 47% cohort rate (+4 pts, n=697).

Weaknesses

Critical risk factors

No factor reached critical level.

Needs improvement

None on this horizon.

Borderline - watch

Between the typical failing and typical passing student. These slide easily.

- Receiving tutoring - 0 = No. 44% of students at this level pass, against a 47% cohort rate (-2 pts, n=1,671).

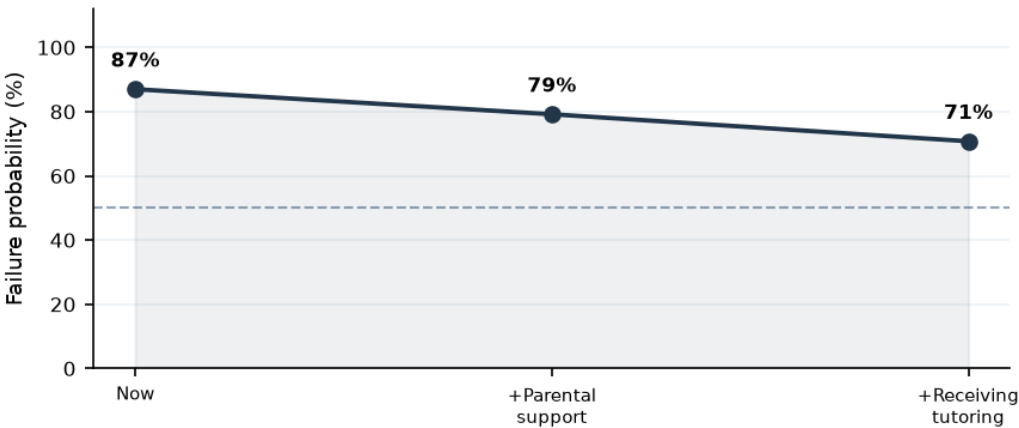
Background factors raising this student's modelled risk: Family income band, Age, Gender, Mother's education, Course topic. The school cannot change these and they must never be used to judge the student or lower expectations - they indicate who needs more resourcing.

E3 - recommendations

Each row was produced by changing the value and re-running the model. Targets come from what passing students in the same source study actually look like. Apply in order - each figure already assumes the ones above it.

#	Change	Now	Target	Failure risk	Alone	Status
1	Parental support	3 = High	4 = Very high	87% to 79%	8 pts on its own	Good
2	Receiving tutoring	0 = No	1 = Yes	79% to 71%	6 pts on its own	Moderate

Combined effect: doing all 2 takes the failure probability from 87% to 71%.



Immediate actions

Things to start this week.

- Give parents one concrete nightly task - check the planner - not 'be supportive'.
- Make the first call home a positive one.
- Arrange tutoring now - small groups of 3-5 are close to 1:1 in effect and far cheaper.
- Brief the tutor with this report's factor list so sessions target the real gap.

Suggestions (longer term)

Sustained changes that hold the improvement in place. These come from professional practice rather than the model.

- Weekly one-page progress summary parents can act on.
- Where home support genuinely is not available, substitute a school mentor.
- Keep tutoring running once grades recover; withdrawing it early undoes the gain.
- Tutor works on this week's class content, not a parallel syllabus.

Intervention priority

When	What
Immediate (this week)	- Parental support: change from 3 = High to 4 = Very high - Receiving tutoring: change from 0 = No to 1 = Yes
Within one month	- Re-run this report after the next assessment.
Long term (this term and beyond)	- Weekly one-page progress summary parents can act on. - Where home support genuinely is not available, substitute a school mentor. - Keep tutoring running once grades recover; withdrawing it early undoes the gain.

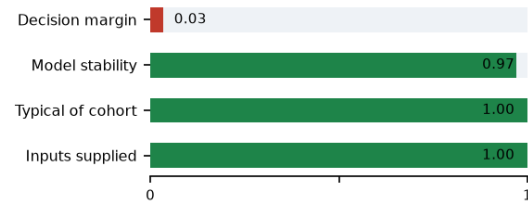
E2 - Early model

Mid-term - background plus engagement and behaviour.

AT RISK - 51.7% fail / 48.3% pass (risk HIGH)

Confidence Moderate (60%). Likely range 51-53%.

This student sits almost exactly on the pass/fail boundary, so the verdict could flip on a small change. Treat it as uncertain, not as a decision.

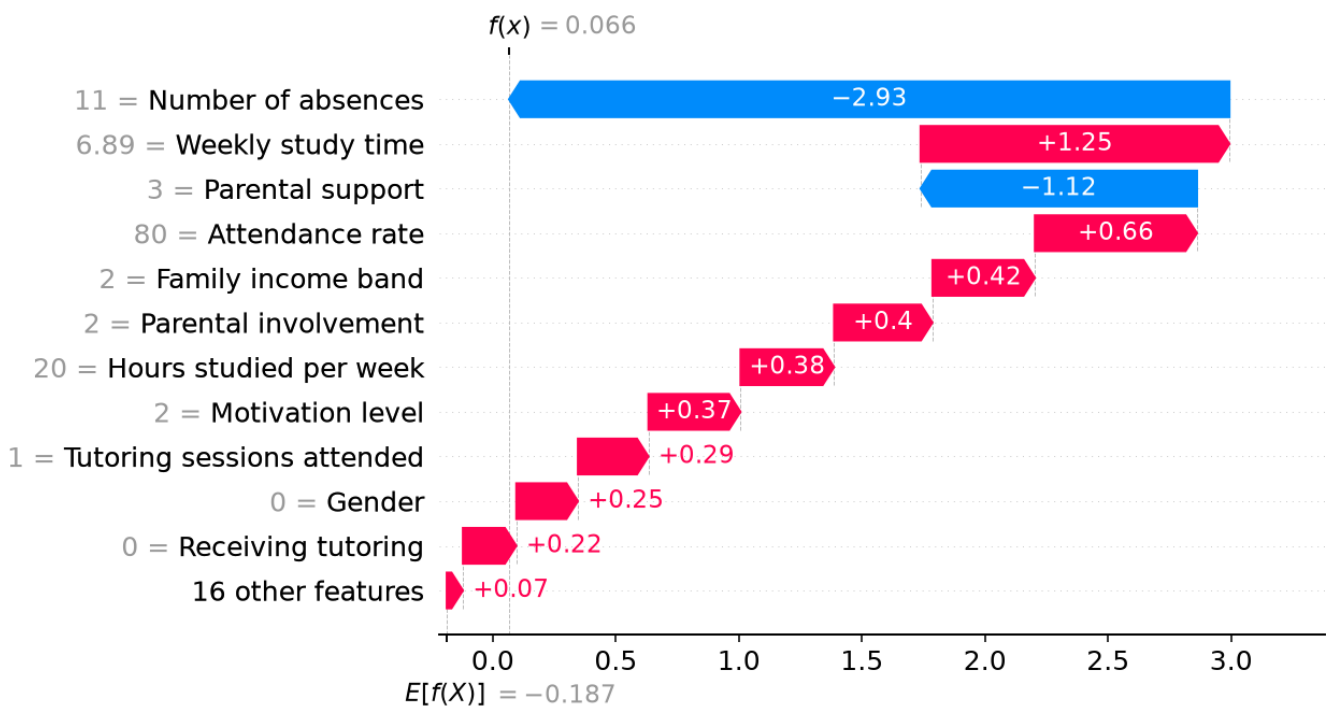


Predicted AT RISK of failing - 52% failure probability (48% chance of passing). Risk level HIGH. Main drivers: Weekly study time; Attendance rate; Family income band. Confidence: Moderate (60/100); likely range 51%-53%. Highest-impact change: Number of absences: move from 11 to 7 classes.

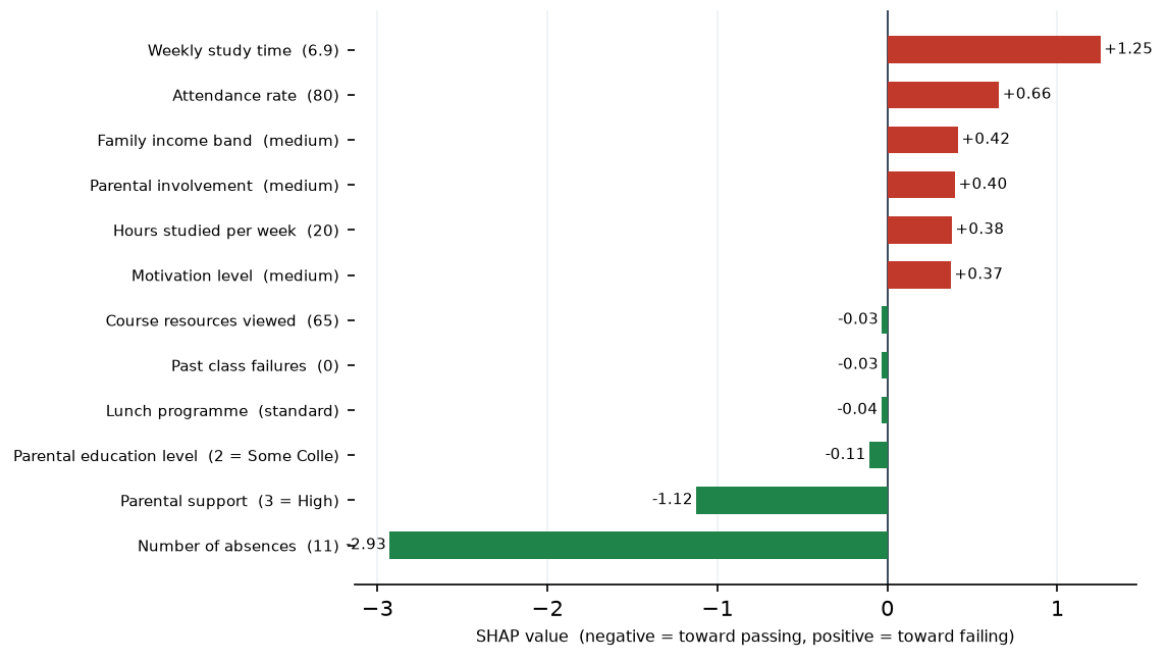
SHAP explainability

Each bar is one answer pushing the prediction away from the cohort average. Red pushes toward failing, blue toward passing; length is the size of the push.

E2 (Early) - how each answer moved the prediction



E2 - top contributors



Read it like this: Weekly study time contributed +1.25 toward predicting failure for this student.

E2 - student profile analysis

Status compares this student against students in the same source study who actually passed. Influence is how strongly the model used the answer for this student. The two can disagree: a good value can still be the model's main worry if everything else is stronger.

Feature	Value	Ideal	Status	Influence	Effect on this prediction
Number of absences	11	3 or below	Moderate	High	lowers risk (-2.93)
Weekly study time	6.9	15.06 or above	Needs improvement	High	raises risk (+1.25)
Parental support	3 = High	4 = Very high	Good	High	lowers risk (-1.12)
Attendance rate	80	-	Not recorded	High	raises risk (+0.66)
Family income band (background)	medium	-	Background	Medium	raises risk (+0.42)
Parental involvement	medium	-	Not recorded	Medium	raises risk (+0.40)
Hours studied per week	20	-	Not recorded	Medium	raises risk (+0.38)
Motivation level	medium	-	Not recorded	Medium	raises risk (+0.37)
Tutoring sessions attended	1	-	Not recorded	Medium	raises risk (+0.29)
Gender (background)	female	male	Background	Medium	raises risk (+0.25)
Receiving tutoring	0 = No	1 = Yes	Moderate	Medium	raises risk (+0.22)
Sleep hours per night	7	-	Not recorded	Medium	raises risk (+0.19)
Parental education level (background)	2 = Some College	1 = High School	Background	Low	lowers risk (-0.11)
Discussion group participation	39	-	Not recorded	Low	raises risk (+0.04)
Lunch programme (background)	standard	-	Background	Low	lowers risk (-0.04)
Test preparation course	none	-	Not recorded	Low	raises risk (+0.03)
Past class failures	0	-	Not recorded	Low	lowers risk (-0.03)
Course resources viewed	65	-	Not recorded	Low	lowers risk (-0.03)
Class participation (hands raised)	50	-	Not recorded	Low	lowers risk (-0.03)
Mother's education (background)	2 = 5th to 9th grade	-	Background	Low	raises risk (+0.02)
Age (background)	15	15 or below	Background	Low	lowers risk (-0.02)
Internet access at home	yes	-	Not recorded	Low	raises risk (+0.01)
Course topic (background)	it	-	Background	Low	raises risk (+0.01)
Announcements viewed	39	-	Not recorded	Low	raises risk (+0.01)
Education stage (background)	middleschool	-	Background	Low	raises risk (+0.00)
Father's education (background)	2 = 5th to 9th grade	-	Background	Low	raises risk (+0.00)
Absence days band	under-7	-	Not recorded	Low	raises risk (+0.00)

Strengths

Excellent areas

None on this horizon.

Good areas

At or above the typical passing student.

- Parental support - 3 = High. 51% of students at this level pass, against a 47% cohort rate (+4 pts, n=697).

Weaknesses

Critical risk factors

No factor reached critical level.

Needs improvement

In the range where students in this dataset usually fail.

- Weekly study time - 6.9. 35th percentile of its cohort; students who pass typically reach 10.55.

Borderline - watch

Between the typical failing and typical passing student. These slide easily.

- Receiving tutoring - 0 = No. 44% of students at this level pass, against a 47% cohort rate (-2 pts, n=1,671).
- Number of absences - 11. 38th percentile of its cohort; students who pass typically reach 7.

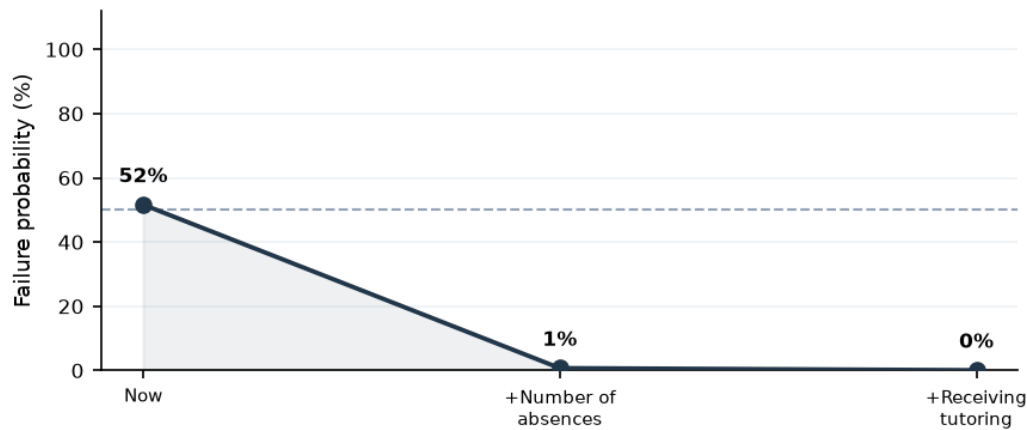
Background factors raising this student's modelled risk: Family income band, Gender, Mother's education, Course topic, Education stage. The school cannot change these and they must never be used to judge the student or lower expectations - they indicate who needs more resourcing.

E2 - recommendations

Each row was produced by changing the value and re-running the model. Targets come from what passing students in the same source study actually look like. Apply in order - each figure already assumes the ones above it.

#	Change	Now	Target	Failure risk	Alone	Status
1	Number of absences	11	7	52% to 1%	51 pts on its own	Moderate
2	Receiving tutoring	0 = No	1 = Yes	1% to 0%	44 pts on its own	Moderate

Combined effect: doing all 2 takes the failure probability from 52% to 0%.



Immediate actions

Things to start this week.

- Call home after the 3rd absence, not the 10th.
- Same-day catch-up materials so returning does not feel hopeless.
- Arrange tutoring now - small groups of 3-5 are close to 1:1 in effect and far cheaper.
- Brief the tutor with this report's factor list so sessions target the real gap.
- Timetable a fixed slot: same time, same place, four days a week.
- Start with two 45-minute focused blocks rather than one long session.

Suggestions (longer term)

Sustained changes that hold the improvement in place. These come from professional practice rather than the model.

- Attendance contract reviewed weekly with the student and family.
- Fix the practical barrier first; motivation conversations come second.
- Keep tutoring running once grades recover; withdrawing it early undoes the gain.
- Tutor works on this week's class content, not a parallel syllabus.
- Teach one technique properly - retrieval practice beats re-reading.
- Offer supervised study at school for students with no quiet space at home.

Intervention priority

When	What
Immediate (this week)	- Number of absences: move from 11 to 7 classes - Receiving tutoring: change from 0 = No to 1 = Yes
Within one month	- Re-run this report after the next assessment.
Long term (this term and beyond)	- Attendance contract reviewed weekly with the student and family. - Fix the practical barrier first; motivation conversations come second. - Keep tutoring running once grades recover; withdrawing it early undoes the gain.

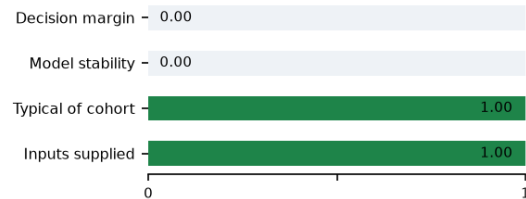
E1 - Late model

Late term - everything above plus actual grades.

AT RISK - 50.0% fail / 50.0% pass (risk HIGH)

Confidence Low (30%). Likely range 47-53%.

This student sits almost exactly on the pass/fail boundary, so the verdict could flip on a small change. Treat it as uncertain, not as a decision.

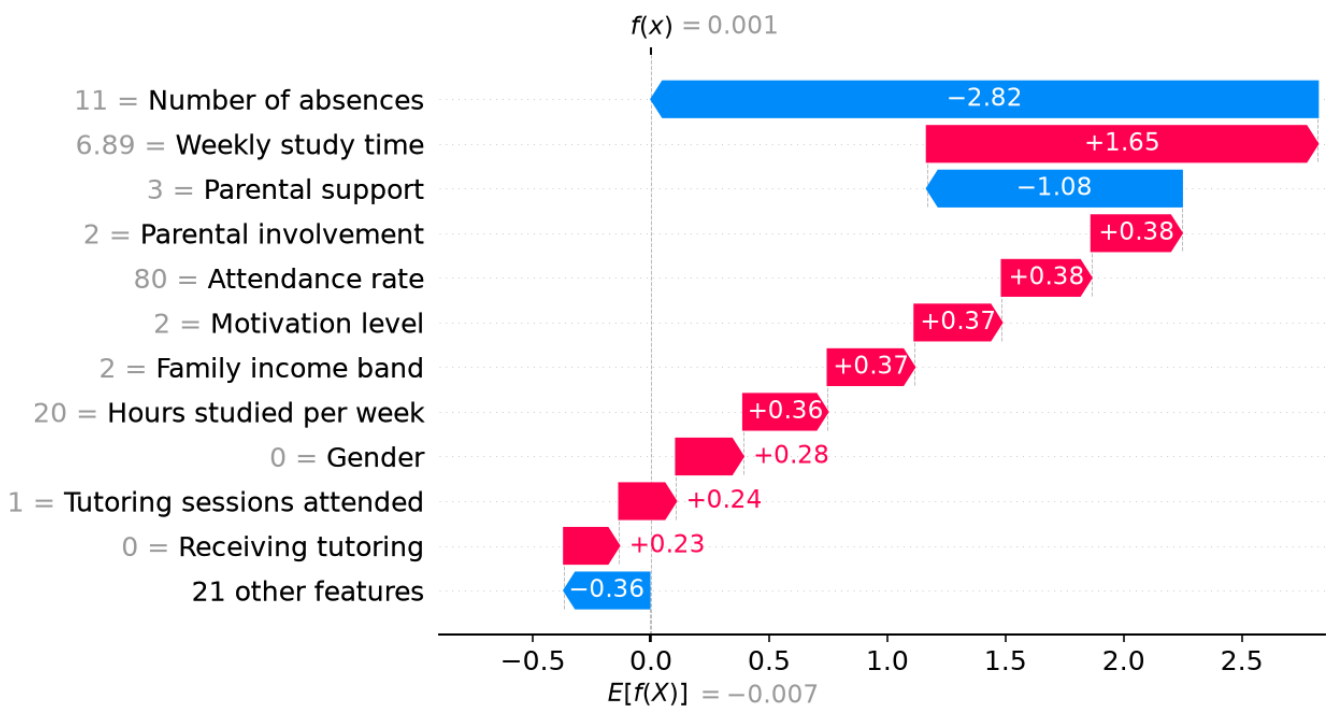


Predicted AT RISK of failing - 50% failure probability (50% chance of passing). Risk level HIGH. Main drivers: Weekly study time; Parental involvement; Attendance rate. Confidence: Low (30/100); likely range 47%-53%. Highest-impact change: Number of absences: move from 11 to 7 classes.

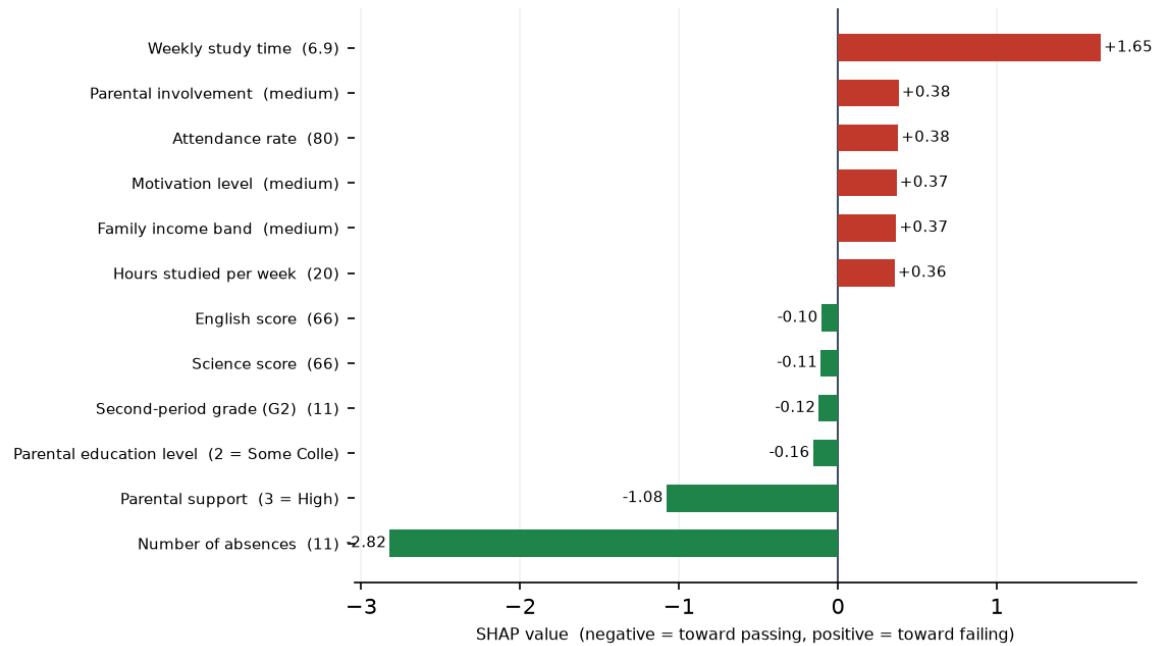
SHAP explainability

Each bar is one answer pushing the prediction away from the cohort average. Red pushes toward failing, blue toward passing; length is the size of the push.

E1 (Late) - how each answer moved the prediction



E1 - top contributors



Read it like this: Weekly study time contributed +1.65 toward predicting failure for this student.

E1 - student profile analysis

Status compares this student against students in the same source study who actually passed. Influence is how strongly the model used the answer for this student. The two can disagree: a good value can still be the model's main worry if everything else is stronger.

Feature	Value	Ideal	Status	Influence	Effect on this prediction
Number of absences	11	3 or below	Moderate	High	lowers risk (-2.82)
Weekly study time	6.9	15.06 or above	Needs improvement	High	raises risk (+1.65)
Parental support	3 = High	4 = Very high	Good	High	lowers risk (-1.08)
Parental involvement	medium	-	Not recorded	Medium	raises risk (+0.38)
Attendance rate	80	-	Not recorded	Medium	raises risk (+0.38)
Motivation level	medium	-	Not recorded	Medium	raises risk (+0.37)
Family income band (background)	medium	-	Background	Medium	raises risk (+0.37)
Hours studied per week	20	-	Not recorded	Medium	raises risk (+0.36)
Gender (background)	female	male	Background	Medium	raises risk (+0.28)
Tutoring sessions attended	1	-	Not recorded	Medium	raises risk (+0.24)
Receiving tutoring	0 = No	1 = Yes	Moderate	Medium	raises risk (+0.23)
Sleep hours per night	7	-	Not recorded	Medium	raises risk (+0.19)
Parental education level (background)	2 = Some College	1 = High School	Background	Medium	lowers risk (-0.16)
Second-period grade (G2)	11	-	Not recorded	Low	lowers risk (-0.12)
Science score	66	-	Not recorded	Low	lowers risk (-0.11)
English score	66	-	Not recorded	Low	lowers risk (-0.10)
Math score	66	-	Not recorded	Low	lowers risk (-0.07)
Course resources viewed	65	-	Not recorded	Low	lowers risk (-0.05)
Discussion group participation	39	-	Not recorded	Low	raises risk (+0.04)
Class participation (hands raised)	50	-	Not recorded	Low	lowers risk (-0.03)
Mother's education (background)	2 = 5th to 9th grade	-	Background	Low	raises risk (+0.03)
Father's education (background)	2 = 5th to 9th grade	-	Background	Low	raises risk (+0.02)
First-period grade (G1)	11	-	Not recorded	Low	lowers risk (-0.02)
Internet access at home	yes	-	Not recorded	Low	raises risk (+0.01)
Course topic (background)	it	-	Background	Low	raises risk (+0.01)
Test preparation course	none	-	Not recorded	Low	raises risk (+0.01)
Age (background)	15	15 or below	Background	Low	lowers risk (-0.00)
Lunch programme (background)	standard	-	Background	Low	raises risk (+0.00)
Past class failures	0	-	Not recorded	Low	raises risk (+0.00)
Education stage (background)	middleschool	-	Background	Low	raises risk (+0.00)
Absence days band	under-7	-	Not recorded	Low	lowers risk (-0.00)
Announcements viewed	39	-	Not recorded	Low	no effect (+0.00)

Strengths

Excellent areas

None on this horizon.

Good areas

At or above the typical passing student.

- Parental support - 3 = High. 51% of students at this level pass, against a 47% cohort rate (+4 pts, n=697).

Weaknesses

Critical risk factors

No factor reached critical level.

Needs improvement

In the range where students in this dataset usually fail.

- Weekly study time - 6.9. 35th percentile of its cohort; students who pass typically reach 10.55.

Borderline - watch

Between the typical failing and typical passing student. These slide easily.

- Receiving tutoring - 0 = No. 44% of students at this level pass, against a 47% cohort rate (-2 pts, n=1,671).
- Number of absences - 11. 38th percentile of its cohort; students who pass typically reach 7.

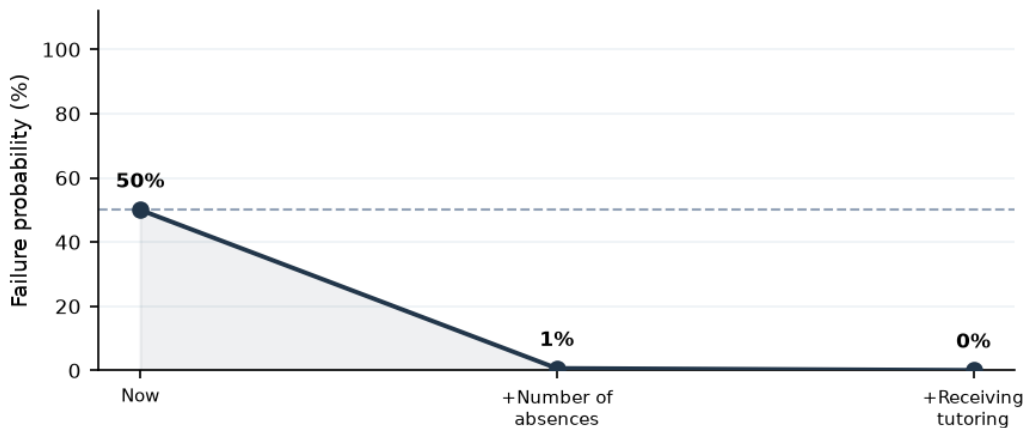
Background factors raising this student's modelled risk: Family income band, Gender, Mother's education, Father's education, Course topic. The school cannot change these and they must never be used to judge the student or lower expectations - they indicate who needs more resourcing.

E1 - recommendations

Each row was produced by changing the value and re-running the model. Targets come from what passing students in the same source study actually look like. Apply in order - each figure already assumes the ones above it.

#	Change	Now	Target	Failure risk	Alone	Status
1	Number of absences	11	7	50% to 1%	49 pts on its own	Moderate
2	Receiving tutoring	0 = No	1 = Yes	1% to 0%	43 pts on its own	Moderate

Combined effect: doing all 2 takes the failure probability from 50% to 0%.



Immediate actions

Things to start this week.

- Call home after the 3rd absence, not the 10th.
- Same-day catch-up materials so returning does not feel hopeless.
- Arrange tutoring now - small groups of 3-5 are close to 1:1 in effect and far cheaper.
- Brief the tutor with this report's factor list so sessions target the real gap.
- Timetable a fixed slot: same time, same place, four days a week.
- Start with two 45-minute focused blocks rather than one long session.

Suggestions (longer term)

Sustained changes that hold the improvement in place. These come from professional practice rather than the model.

- Attendance contract reviewed weekly with the student and family.
- Fix the practical barrier first; motivation conversations come second.
- Keep tutoring running once grades recover; withdrawing it early undoes the gain.
- Tutor works on this week's class content, not a parallel syllabus.
- Teach one technique properly - retrieval practice beats re-reading.
- Offer supervised study at school for students with no quiet space at home.

Intervention priority

When	What
Immediate (this week)	- Number of absences: move from 11 to 7 classes - Receiving tutoring: change from 0 = No to 1 = Yes
Within one month	- Re-run this report after the next assessment.
Long term (this term and beyond)	- Attendance contract reviewed weekly with the student and family. - Fix the practical barrier first; motivation conversations come second. - Keep tutoring running once grades recover; withdrawing it early undoes the gain.

Final summary

The same result, written for three different readers.

For the teacher

Student row 0 carries a 50% modelled probability of failing (risk level HIGH, confidence Low). Strongest areas: Parental support. Weakest: Weekly study time. Intervention priority is within one month. Put a support plan in place this month and set a review date. The modelled action plan would move the failure probability from 50% to <1%.

For the parent or guardian

On the information provided, there is roughly a 50 in 100 chance of not passing this term and a 50 in 100 chance of passing. What is going well: Parental support. What needs attention: Weekly study time. The single change the model responds to most is number of absences: moving from 11 to 7. Please talk to the school about how to support that.

For the student

Right now the model puts your chance of passing at about 50%. That is a prediction from data, not a decision about you, and it can change. You are doing well on: Parental support. What is holding you back: Weekly study time. The thing that would shift this most is number of absences: from 11 to 7. If you and the school work through the full plan, the model's estimate moves to >99% chance of passing.

Appendix A - what each question means

Every value range below is computed from the training data itself, comparing students who passed with students who failed inside the same source study. They describe this dataset, not universal standards.

Sleep hours per night (*sleep_hours*)

Average nightly sleep. Under-sleeping damages memory consolidation, so study hours convert poorly into recall.

Possible values: 4 to 10 hours / night

Only recorded in the StudentFactors study, so it is graded against those 6,607 students.

Value	Interpretation	Meaning in this cohort
4	Neutral	99% of these students pass (+0 pts vs the 99% cohort rate), n=309
5	Neutral	99% of these students pass (-0 pts vs the 99% cohort rate), n=695
6	Neutral	99% of these students pass (+0 pts vs the 99% cohort rate), n=1,376
7	Neutral	99% of these students pass (-0 pts vs the 99% cohort rate), n=1,741
8	Neutral	99% of these students pass (+0 pts vs the 99% cohort rate), n=1,399
9	Neutral	99% of these students pass (-0 pts vs the 99% cohort rate), n=775
10	Neutral	99% of these students pass (-0 pts vs the 99% cohort rate), n=312

Parental support (*parental_support*)

Rated support at home, coded 0-4 from none to very high. Covers practical help (a quiet space, checking homework) as well as encouragement.

Possible values: 0 = None 1 = Low 2 = Moderate 3 = High 4 = Very high

Only recorded in the StudentPerf study, so it is graded against those 2,392 students.

Value	Interpretation	Meaning in this cohort
0 = None	Critical	36% of these students pass (-11 pts vs the 47% cohort rate), n=212
1 = Low	Needs improvement	42% of these students pass (-5 pts vs the 47% cohort rate), n=489
2 = Moderate	Moderate	46% of these students pass (-1 pts vs the 47% cohort rate), n=740
3 = High	Good	51% of these students pass (+4 pts vs the 47% cohort rate), n=697
4 = Very high	Good	56% of these students pass (+9 pts vs the 47% cohort rate), n=254

Parental involvement (*parental_involvement*)

Rated parent engagement with the student's education.

Possible values: One of: low, medium, high

Only recorded in the StudentFactors study, so it is graded against those 6,607 students.

Value	Interpretation	Meaning in this cohort
low	Moderate	98% of these students pass (-1 pts vs the 99% cohort rate), n=1,337
medium	Moderate	99% of these students pass (+0 pts vs the 99% cohort rate), n=3,362
high	Moderate	99% of these students pass (+1 pts vs the 99% cohort rate), n=1,908

Family income band (*family_income*) **background - not actionable**

Household income band. Background context - it flags which students need resources provided, never which students to expect less from.

Possible values: One of: low, medium, high

Only recorded in the StudentFactors study, so it is graded against those 6,607 students.

Value	Interpretation	Meaning in this cohort
low	Moderate	99% of these students pass (-0 pts vs the 99% cohort rate), n=2,672
medium	Moderate	99% of these students pass (+0 pts vs the 99% cohort rate), n=2,666
high	Moderate	100% of these students pass (+1 pts vs the 99% cohort rate), n=1,269

Age (*age*) **background - not actionable**

Age in years. Background context; older than the cohort often indicates a previously repeated year.

Possible values: 10 to 22 years (cohort median 16)

Only recorded in the StudentGrades, StudentPerf, xAPI study, so it is graded against those 3,916 students.

Value	Interpretation	Meaning in this cohort
15 and below	Excellent	better than most students who passed
15 to 16	Good	at or below the typical passing student
16 to 16	Moderate	between the typical passing and failing student
16 to 18	Needs improvement	in the range where students usually fail
above 18	Critical	worse than 75% of students who failed

Motivation level (*motivation_level*)

Rated engagement with schooling. Low motivation is usually a symptom - of repeated failure or of having no clear goal - rather than a root cause.

Possible values: One of: low, medium, high

Only recorded in the StudentFactors study, so it is graded against those 6,607 students.

Value	Interpretation	Meaning in this cohort
low	Moderate	98% of these students pass (-1 pts vs the 99% cohort rate), n=1,937
medium	Moderate	99% of these students pass (+0 pts vs the 99% cohort rate), n=3,351
high	Moderate	99% of these students pass (+0 pts vs the 99% cohort rate), n=1,319

Gender (*gender*) **background - not actionable**

Recorded gender. Background context only, never a target for intervention.

Possible values: One of: female, male

Value	Interpretation	Meaning in this cohort
female	Neutral	83% of these students pass (-0 pts vs the 83% cohort rate), n=5,247
male	Neutral	84% of these students pass (+0 pts vs the 83% cohort rate), n=6,276

Receiving tutoring (*tutoring*)

Whether the student currently receives tutoring support. Important caveat: tutoring is usually GIVEN TO students who are already struggling, so in raw data it can look associated with failure. Compared fairly against similar students, tutoring is associated with a higher pass rate.

Possible values: 0 = No 1 = Yes

Only recorded in the StudentPerf study, so it is graded against those 2,392 students.

Value	Interpretation	Meaning in this cohort
0 = No	Moderate	44% of these students pass (-2 pts vs the 47% cohort rate), n=1,671
1 = Yes	Good	52% of these students pass (+5 pts vs the 47% cohort rate), n=721

Mother's education (*Medu*) *background - not actionable*

Mother's education, coded 0-4 on the UCI scale. Background context only.

Possible values: 0 = None 1 = Primary (4th grade) 2 = 5th to 9th grade 3 = Secondary education 4 = Higher education

Only recorded in the StudentGrades study, so it is graded against those 1,044 students.

Value	Interpretation	Meaning in this cohort
0 = None	Moderate	89% of these students pass (+11 pts vs the 78% cohort rate), n=9
1 = Primary (4th grade)	Needs improvement	69% of these students pass (-9 pts vs the 78% cohort rate), n=202
2 = 5th to 9th grade	Moderate	78% of these students pass (+0 pts vs the 78% cohort rate), n=289
3 = Secondary education	Moderate	76% of these students pass (-2 pts vs the 78% cohort rate), n=238
4 = Higher education	Good	85% of these students pass (+7 pts vs the 78% cohort rate), n=306

Lunch programme (*lunch*) *background - not actionable*

Lunch programme status, widely used as a proxy for household income. Background context - it identifies who needs resourcing.

Possible values: One of: free/reduced, standard

Only recorded in the PerfScores study, so it is graded against those 1,000 students.

Value	Interpretation	Meaning in this cohort
free/reduced	Critical	56% of these students pass (-15 pts vs the 72% cohort rate), n=355
standard	Good	80% of these students pass (+8 pts vs the 72% cohort rate), n=645

Discussion group participation (*discussion*)

Count of contributions to discussion boards. Explaining an idea to a peer is one of the strongest retention activities available.

Possible values: 1 to 99 posts this term (cohort median 39)

Only recorded in the xAPI study, so it is graded against those 480 students.

Value	Interpretation	Meaning in this cohort
70 and above	Excellent	top quartile of students who passed
43 to 70	Good	at or above the typical passing student
13 to 43	Moderate	between the typical failing and passing student
7 to 13	Needs improvement	in the range where students usually fail
below 7	Critical	worse than 75% of students who failed

Test preparation course (*test_prep*)

Whether the student completed a structured exam-preparation programme.

Possible values: One of: none, completed

Only recorded in the PerfScores study, so it is graded against those 1,000 students.

Value	Interpretation	Meaning in this cohort
none	Needs improvement	65% of these students pass (-7 pts vs the 72% cohort rate), n=642
completed	Excellent	83% of these students pass (+12 pts vs the 72% cohort rate), n=358

Internet access at home (*internet_access*)

Whether the student can get online at home. Without it, every online task becomes a school-hours-only task.

Possible values: One of: no, yes

Only recorded in the StudentFactors study, so it is graded against those 6,607 students.

Value	Interpretation	Meaning in this cohort
no	Moderate	98% of these students pass (-1 pts vs the 99% cohort rate), n=499
yes	Moderate	99% of these students pass (+0 pts vs the 99% cohort rate), n=6,108

Parental education level (*parental_education*) **background - not actionable**

Highest parental education, coded 0-4. Background context. It tells you how much explanation a family may need from the school - it says nothing about what the student is capable of.

Possible values: 0 = None 1 = High School 2 = Some College 3 = Bachelor's degree 4 = Postgraduate

Only recorded in the StudentPerf study, so it is graded against those 2,392 students.

Value	Interpretation	Meaning in this cohort
0 = None	Moderate	46% of these students pass (-1 pts vs the 47% cohort rate), n=243
1 = High School	Moderate	48% of these students pass (+2 pts vs the 47% cohort rate), n=728
2 = Some College	Moderate	47% of these students pass (+1 pts vs the 47% cohort rate), n=934
3 = Bachelor's degree	Moderate	45% of these students pass (-2 pts vs the 47% cohort rate), n=367
4 = Postgraduate	Needs improvement	39% of these students pass (-8 pts vs the 47% cohort rate), n=120

Past class failures (*failures*)

Count of previously failed classes. A prior failure usually signals an unclosed knowledge gap that keeps breaking later topics.

Possible values: 0 to 3 classes

Only recorded in the StudentGrades study, so it is graded against those 1,044 students.

Value	Interpretation	Meaning in this cohort
0	Good	85% of these students pass (+7 pts vs the 78% cohort rate), n=861
1	Critical	52% of these students pass (-26 pts vs the 78% cohort rate), n=120
2	Critical	33% of these students pass (-45 pts vs the 78% cohort rate), n=33
3	Critical	30% of these students pass (-48 pts vs the 78% cohort rate), n=30

Course resources viewed (*visited_resources*)

Count of accesses to course material in the learning platform. Read it together with grades: a low count with good grades means the student works offline; a low count with weak grades usually means little studying at all.

Possible values: 0 to 99 views this term (cohort median 65)

Only recorded in the xAPI study, so it is graded against those 480 students.

Value	Interpretation	Meaning in this cohort
87 and above	Excellent	top quartile of students who passed
75 to 87	Good	at or above the typical passing student
8 to 75	Moderate	between the typical failing and passing student
5 to 8	Needs improvement	in the range where students usually fail
below 5	Critical	worse than 75% of students who failed

Class participation (hands raised) (*raisedhands*)

Number of times the student volunteered an answer or question in class this term, logged by the teacher. It is a proxy for classroom engagement. A quiet student is not automatically disengaged, but a count near zero almost always means the student has stopped following lessons.

Possible values: 0 to 100 times this term (cohort median 50)

Only recorded in the xAPI study, so it is graded against those 480 students.

Value	Interpretation	Meaning in this cohort
79 and above	Excellent	top quartile of students who passed
60 to 79	Good	at or above the typical passing student
10 to 60	Moderate	between the typical failing and passing student
5 to 10	Needs improvement	in the range where students usually fail
below 5	Critical	worse than 75% of students who failed

Announcements viewed (*announcements_view*)

Count of announcement views. Low numbers usually mean missed deadlines rather than laziness - the student never saw the message.

Possible values: 0 to 98 views this term (cohort median 33)

Only recorded in the xAPI study, so it is graded against those 480 students.

Value	Interpretation	Meaning in this cohort
62 and above	Excellent	top quartile of students who passed
40 to 62	Good	at or above the typical passing student
10 to 40	Moderate	between the typical failing and passing student
4 to 10	Needs improvement	in the range where students usually fail
below 4	Critical	worse than 75% of students who failed

Course topic (topic) *background - not actionable*

Subject the student is enrolled in. Useful for spotting subjects with systematically higher failure rates, not for judging the student.

Possible values: One of: arabic, biology, chemistry, english, french, geology, history, it, math, quran, science, spanish

Only recorded in the xAPI study, so it is graded against those 480 students.

Value	Interpretation	Meaning in this cohort
arabic	Moderate	92% of these students pass (+7 pts vs the 85% cohort rate), n=59
biology	Moderate	87% of these students pass (+2 pts vs the 85% cohort rate), n=30
chemistry	Moderate	88% of these students pass (+3 pts vs the 85% cohort rate), n=24
english	Moderate	89% of these students pass (+4 pts vs the 85% cohort rate), n=45
french	Moderate	83% of these students pass (-2 pts vs the 85% cohort rate), n=65
geology	Excellent	100% of these students pass (+15 pts vs the 85% cohort rate), n=24
history	Moderate	95% of these students pass (+10 pts vs the 85% cohort rate), n=19
it	Critical	73% of these students pass (-12 pts vs the 85% cohort rate), n=95
math	Critical	71% of these students pass (-13 pts vs the 85% cohort rate), n=21
quran	Moderate	91% of these students pass (+6 pts vs the 85% cohort rate), n=22
science	Moderate	90% of these students pass (+6 pts vs the 85% cohort rate), n=51
spanish	Moderate	76% of these students pass (-9 pts vs the 85% cohort rate), n=25

Father's education (Fedu) *background - not actionable*

Father's education, coded 0-4 on the UCI scale. Background context only.

Possible values: 0 = None 1 = Primary (4th grade) 2 = 5th to 9th grade 3 = Secondary education 4 = Higher education

Only recorded in the StudentGrades study, so it is graded against those 1,044 students.

Value	Interpretation	Meaning in this cohort
0 = None	Excellent	100% of these students pass (+22 pts vs the 78% cohort rate), n=9
1 = Primary (4th grade)	Needs improvement	69% of these students pass (-9 pts vs the 78% cohort rate), n=256
2 = 5th to 9th grade	Moderate	79% of these students pass (+1 pts vs the 78% cohort rate), n=324
3 = Secondary education	Moderate	81% of these students pass (+3 pts vs the 78% cohort rate), n=231
4 = Higher education	Good	83% of these students pass (+5 pts vs the 78% cohort rate), n=224

Education stage (stage) *background - not actionable*

School stage (lower level, middle school, high school). Structural context rather than something to act on.

Possible values: One of: lowerlevel, middleschool, highschool

Only recorded in the xAPI study, so it is graded against those 480 students.

Value	Interpretation	Meaning in this cohort
lowerlevel	Neutral	81% of these students pass (-3 pts vs the 85% cohort rate), n=199
middleschool	Neutral	87% of these students pass (+3 pts vs the 85% cohort rate), n=248
highschool	Neutral	85% of these students pass (+0 pts vs the 85% cohort rate), n=33

Absence days band (*absence_days*)

Banded absence count recorded in the xAPI study: either under 7 days or above 7 days. Crossing 7 days is where attendance stops being incidental.

Possible values: One of: under-7, above-7

Only recorded in the xAPI study, so it is graded against those 480 students.

Value	Interpretation	Meaning in this cohort
under-7	Excellent	95% of these students pass (+10 pts vs the 85% cohort rate), n=289
above-7	Critical	69% of these students pass (-15 pts vs the 85% cohort rate), n=191

Number of absences (*absences*)

Total classes missed. Across almost every education dataset this is the single strongest behavioural predictor of failure - each missed class is content the student must recover alone.

Possible values: 0 to 29 classes (cohort median 15)

Only recorded in the StudentPerf study, so it is graded against those 2,392 students.

Value	Interpretation	Meaning in this cohort
3 and below	Excellent	better than most students who passed
3 to 7	Good	at or below the typical passing student
7 to 21	Moderate	between the typical passing and failing student
21 to 25	Needs improvement	in the range where students usually fail
above 25	Critical	worse than 75% of students who failed

Weekly study time (*study_time*)

Self-reported weekly study hours outside lessons. Beyond the cohort's upper quartile the returns flatten - how the time is used matters more than how much of it there is.

Possible values: 0.0 to 20.0 hours / week (cohort median 9.705)

Only recorded in the StudentPerf study, so it is graded against those 2,392 students.

Value	Interpretation	Meaning in this cohort
15.1 and above	Excellent	top quartile of students who passed
10.6 to 15.1	Good	at or above the typical passing student
8.8 to 10.6	Moderate	between the typical failing and passing student
4.3 to 8.8	Needs improvement	in the range where students usually fail
below 4.3	Critical	worse than 75% of students who failed

Attendance rate (*attendance*)

Share of scheduled classes attended. Below roughly 90% students start missing whole topics rather than single lessons.

Possible values: 60 to 100 % (cohort median 80)

Only recorded in the StudentFactors study, so it is graded against those 6,607 students.

Value	Interpretation	Meaning in this cohort
90 and above	Excellent	top quartile of students who passed
80 to 90	Good	at or above the typical passing student
63 to 80	Moderate	between the typical failing and passing student
61 to 63	Needs improvement	in the range where students usually fail
below 61	Critical	worse than 75% of students who failed

Hours studied per week (*hours_studied*)

Weekly self-study hours outside lessons.

Possible values: 1 to 44 hours / week (cohort median 20)

Only recorded in the StudentFactors study, so it is graded against those 6,607 students.

Value	Interpretation	Meaning in this cohort
24 and above	Excellent	top quartile of students who passed
20 to 24	Good	at or above the typical passing student
11 to 20	Moderate	between the typical failing and passing student
7 to 11	Needs improvement	in the range where students usually fail
below 7	Critical	worse than 75% of students who failed

Tutoring sessions attended (*tutoring_sessions*)

Number of tutoring sessions attended. Frequency matters more than session length - weekly contact stops gaps compounding. As with the tutoring flag, sessions are usually allocated to students already behind, so compare against similar students rather than the whole cohort.

Possible values: 0 to 8 sessions

Only recorded in the StudentFactors study, so it is graded against those 6,607 students.

Value	Interpretation	Meaning in this cohort
0	Moderate	98% of these students pass (-1 pts vs the 99% cohort rate), n=1,513
1	Moderate	99% of these students pass (+0 pts vs the 99% cohort rate), n=2,179
2	Moderate	100% of these students pass (+1 pts vs the 99% cohort rate), n=1,649
3	Moderate	99% of these students pass (+0 pts vs the 99% cohort rate), n=836
4	Moderate	99% of these students pass (+0 pts vs the 99% cohort rate), n=301
5	Moderate	100% of these students pass (+1 pts vs the 99% cohort rate), n=103
6	Moderate	100% of these students pass (+1 pts vs the 99% cohort rate), n=18
7	Moderate	100% of these students pass (+1 pts vs the 99% cohort rate), n=7
8	Moderate	100% of these students pass (+1 pts vs the 99% cohort rate), n=1

Second-period grade (G2) (*G2*)

Grade from the second assessment period, on the 0-20 scale. Read with G1 it shows the trajectory: improving, flat or sliding.

Possible values: 0 to 19 0-20 (cohort median 11)

Only recorded in the StudentGrades study, so it is graded against those 1,044 students.

Value	Interpretation	Meaning in this cohort
14 and above	Excellent	top quartile of students who passed
12 to 14	Good	at or above the typical passing student
8 to 12	Moderate	between the typical failing and passing student
6 to 8	Needs improvement	in the range where students usually fail
below 6	Critical	worse than 75% of students who failed

Science score (*science_score*)

Science score out of 100.

Possible values: 17 to 100 /100 (cohort median 70)

Only recorded in the PerfScores study, so it is graded against those 1,000 students.

Value	Interpretation	Meaning in this cohort
83 and above	Excellent	top quartile of students who passed
75 to 83	Good	at or above the typical passing student
54 to 75	Moderate	between the typical failing and passing student
48 to 54	Needs improvement	in the range where students usually fail
below 48	Critical	worse than 75% of students who failed

English score (*english_score*)

English / language score out of 100. Weak reading fluency quietly lowers marks in every other written subject.

Possible values: 10 to 100 /100 (cohort median 69)

Only recorded in the PerfScores study, so it is graded against those 1,000 students.

Value	Interpretation	Meaning in this cohort
82 and above	Excellent	top quartile of students who passed
74 to 82	Good	at or above the typical passing student
52 to 74	Moderate	between the typical failing and passing student
45 to 52	Needs improvement	in the range where students usually fail
below 45	Critical	worse than 75% of students who failed

Math score (*math_score*)

Mathematics score out of 100. Weak numeracy tends to pull science and technical subjects down with it.

Possible values: 0 to 100 /100 (cohort median 66)

Only recorded in the PerfScores study, so it is graded against those 1,000 students.

Value	Interpretation	Meaning in this cohort
80 and above	Excellent	top quartile of students who passed
72 to 80	Good	at or above the typical passing student
50 to 72	Moderate	between the typical failing and passing student
44 to 50	Needs improvement	in the range where students usually fail
below 44	Critical	worse than 75% of students who failed

First-period grade (G1) (G1)

Grade from the first assessment period, on the 0-20 scale. The earliest hard academic signal and the cheapest moment to intervene.

Possible values: 0 to 19 0-20 (cohort median 11)

Only recorded in the StudentGrades study, so it is graded against those 1,044 students.

Value	Interpretation	Meaning in this cohort
14 and above	Excellent	top quartile of students who passed
12 to 14	Good	at or above the typical passing student
8 to 12	Moderate	between the typical failing and passing student
7 to 8	Needs improvement	in the range where students usually fail
below 7	Critical	worse than 75% of students who failed

Appendix B - method, data quality and limitations

This dataset merges five studies

The training file stacks several different studies on top of each other: StudentFactors (6,607 rows); StudentPerf (2,392 rows); StudentGrades (1,044 rows); PerfScores (1,000 rows); xAPI (480 rows).

Each study only fills in its own columns. Every other column holds a constant placeholder, which makes whole-file statistics misleading. Tutoring is the clearest example: across the whole file, students with tutoring appear to pass far less often, because 'no tutoring' silently includes thousands of placeholder rows from studies with very high pass rates. Compared only against students from the same study, tutoring is associated with a higher pass rate, which is what you would expect.

Every benchmark, percentile, pass rate and recommendation target in this report is therefore computed inside the study that the column came from. The number of students used for each comparison is stated in Appendix A.

The three horizons

E3 uses only enrolment and background data, so it is available on day one but is the least accurate. E2 adds attendance and engagement. E1 adds grades and is the most accurate but arrives too late to change much. Read them together: E3 says who to watch, E1 says what happened.

Probability and risk level

The failure probability is the model's estimate for a student with this profile. 50% is the decision threshold, but it is a convention: a student at 48% is not meaningfully safer than one at 52%.

Risk levels group the probability into bands - LOW under 20%, MODERATE 20-40%, HIGH 40-70%, CRITICAL above 70% - so that action can be prioritised across a class.

Confidence

A 0-100% score from four measurements. Decision margin: distance from the 50% threshold. Model stability: whether the prediction stayed put as trees were added, and whether any snapshot disagreed with the final verdict. Typicality: whether the answers fall inside the range the model was trained on. Completeness: how many inputs were supplied.

Low confidence does not mean the prediction is wrong. It means it should not be acted on by itself.

Status grades

Excellent, Good, Moderate, Needs improvement and Critical come from comparing the student against the real distribution of passing and failing students in the same source study. Neutral means the factor does not separate the two groups well enough to grade honestly. Background marks fields nobody can act on.

Recommendations versus suggestions

Recommendations are counterfactual simulations: the value is changed, the model is re-run, and the resulting change in failure probability is reported. They are ordered greedily, so each figure already accounts for the ones above it.

Suggestions are practical guidance on how to achieve those changes and come from professional practice, not from the model. The system is built so that these can be replaced by generated, personalised guidance later without changing any of the numbers.

Limitations - read before acting

The model learns associations, not causes. A simulated drop in risk is what the model predicts for a student who already has that value; it is not a guarantee that changing it will produce that outcome.

Some relationships are confounded. Tutoring and tutoring sessions are allocated to students who are already behind, so their raw association with failure reflects who receives them, not what they do. Where the data contradicts the obvious direction, this report leaves the factor ungraded rather than advising something harmful.

Background factors such as gender, income and parental education may carry predictive weight because of inequities in the underlying data. They identify students who need more support. They must never be used to lower expectations of a student.

Predictions describe a probability across similar students, not a destiny for this one. Share the reasoning with the student and let a human make the decision.